Hints: Polynomials – 1. Roots, multiplicity and the derivative

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The roots of $f(2x)$ are $x_1/2, x_2/2, x_3/2$, and f' divides $f(2x)$, so (after renumbering) $f'(x) = 3(x - \frac{x_1}{2})(x - \frac{x_2}{2})$ for monic f.

Hint 2. Compare with $f'(x) = 3x^2 - 2(x_1 + x_2 + x_3)x + (x_1x_2 + x_1x_3 + x_2x_3)$: the sum gives $x_1 + x_2 = -4x_3$, and the product then gives $x_1x_2 = 16x_3^2$.

Hint 3. So x_1, x_2 are the roots of $t^2 + 4x_3t + 16x_3^2$, i.e. $x_{1,2} = 4\omega^{\pm 1}x_3$ with $\omega = e^{2\pi i/3}$. The ratios are $x_1 : x_2 : x_3 = 4\omega : 4\omega^2 : 1$ (apart from the trivial case $f = x^3$).

Problem 2

Hint 1. Angles $A, B, C \in (0, \pi)$ form a triangle iff $A + B + C = \pi$. Find a polynomial identity in $\cos A, \cos B, \cos C$ expressing this.

Hint 2. If $A + B + C = \pi$ then $\cos^2 A + \cos^2 B + \cos^2 C + 2\cos A \cos B \cos C = 1$. With $u + v + w = -p$, $uv + vw + wu = q$, $uvw = -r$ this reads $p^2 - 2q - 2r = 1$.

Hint 3. Sufficiency: if $u, v, w > 0$ satisfy $u^2 + v^2 + w^2 + 2uvw = 1$ then $u, v < 1$; write $u = \cos A$, $v = \cos B$ with $A, B \in (0, \pi/2)$ and solve the quadratic for w: its roots are $-\cos(A + B)$ and $-\cos(A - B) < 0$. So $w = \cos(\pi - A - B)$ and $C = \pi - A - B$ works. Answer: $p^2 - 2q - 2r = 1$.

Problem 3

Hint 1. Since $c^2 = a^2d \neq 0$, we have $a \neq 0$ and $d \neq 0$ (so no root is 0). Put $k = c/a$; then $c = ak$, $d = k^2$, and the equation is $x^4 + ax^3 + bx^2 + akx + k^2 = 0$. It is symmetric under $x \mapsto k/x$: divide by x^2.

Hint 2. With $y = x + \frac{k}{x}$ we get $x^2 + \frac{k^2}{x^2} = y^2 - 2k$, so the equation becomes $y^2 + ay + (b - 2k) = 0$. Equivalently, if y_1, y_2 are the roots of this quadratic, then (expand to check)

$$x^4 + ax^3 + bx^2 + cx + d = (x^2 - y_1x + k)(x^2 - y_2x + k).$$

Hint 3. The two roots of each quadratic factor have product k. Call the roots of the first factor r_1, r_4 and those of the second r_2, r_3: then $r_1r_4 = k = r_2r_3$, i.e. $r_1/r_2 = r_3/r_4$.

Problem 4

Hint 1. The condition $\sum \frac{a_k}{k+1} = 0$ is the value at 1 of an antiderivative of the polynomial.

Hint 2. $F(x) = \sum_k \frac{a_k}{k+1}x^{k+1}$ satisfies $F(0) = F(1) = 0$; apply Rolle's theorem to get a root of F' in $(0, 1)$.

Problem 5

Hint 1. Both recurrences have the form $Q_{n+1} = Q_n + c_n\, x\, Q_{n-1}$ with $c_n > 0$ and $c_n \to \infty$. Note $Q_n(0) = 1$ and all coefficients are positive, so real roots are negative.

Hint 2. Monotonicity by induction: if $x_{n-1} < x_n$ then $Q_{n-1} > 0$ on (x_{n-1}, ∞), so $Q_{n+1}(x_n) = c_n x_n Q_{n-1}(x_n) < 0 < Q_{n+1}(0)$. Hence Q_{n+1} has a real root in $(x_n, 0)$ and $x_{n+1} > x_n$. Start with $x_1 = -1$, $x_2 = -\frac{1}{2}$.

Hint 3. So $x_n \uparrow L \leqslant 0$. If $L < 0$, fix $x \in (L, 0)$: all $Q_n(x) > 0$, and $Q_{n+1}(x) = Q_n(x) - c_n|x|Q_{n-1}(x) < Q_n(x)$. Thus $Q_{n+1}(x) < Q_{n-1}(x)(1 - c_n|x|)$, which is negative once $c_n|x| > 1$, a contradiction.

Problem 6

Hint 1. Try to factor Q as a product of two expressions, each linear in P and P'.

Hint 2. $Q = (P + xP')(xP + P')$, and $P + xP' = (xP)'$, $xP + P' = e^{-x^2/2}(e^{x^2/2}P)'$.

Hint 3. Let $1 < r_1 < \cdots < r_N$ ($N \geqslant n$) be all the real roots of P exceeding 1. Rolle for xP (roots $0, r_1, \ldots, r_N$) gives a root of the first factor in $(0, r_1)$ and in each (r_i, r_{i+1}); Rolle for $e^{x^2/2}P$ gives a root of the second factor in each (r_i, r_{i+1}).

Hint 4. A common root of the two factors satisfies $P' = -xP$ and $P = -xP'$, hence $(1-x^2)P = 0$; in (r_i, r_{i+1}) neither is possible. So the roots found are distinct and there are at least $N + (N-1) \geqslant 2n - 1$ of them: the statement is true.

Problem 7

Hint 1. Build the sequence inductively: given p_n with n distinct real roots, choose a_{n+1} very small.

Hint 2. Pick points $y_0 < y_1 < \cdots < y_n$ separating the roots of p_n (with y_n beyond the largest root); p_n alternates in sign on them. For $|a_{n+1}|$ small enough, $p_{n+1} = p_n + a_{n+1}x^{n+1}$ has the same signs at the y_i.

Hint 3. Choose the sign of a_{n+1} opposite to that of $p_n(y_n)$; then p_{n+1} also changes sign on (y_n, ∞), giving $n + 1$ distinct real roots. Answer: yes.

Problem 8

Hint 1. Rescale: put $y = \frac{a-x}{a} \in (0, 1)$ (note $a > 0$).

Hint 2. The polynomial becomes $\frac{a^6 y^2}{2}(2y^4 - 6y^3 + 5y^2 - 1)$. Check that $y = 1$ is a root of the quartic, in fact a double one.

Hint 3. $2y^4 - 6y^3 + 5y^2 - 1 = (y-1)^2(2y^2 - 2y - 1)$, and $2y^2 - 2y - 1 = 2y(y-1) - 1 < 0$ on $(0, 1)$.

Problem 9

Hint 1. Let $x \to \infty$ to get $|a| \leqslant |A|$; by changing signs assume $A > 0$. Split according to the sign of $B^2 - 4AC$.

Hint 2. If $B^2 - 4AC > 0$, the two distinct real roots of $Ax^2 + Bx + C$ are roots of $ax^2 + bx + c$, so $ax^2 + bx + c = \frac{a}{A}(Ax^2 + Bx + C)$ and the claim follows from $|a| \leqslant A$.

Hint 3. If $B^2 - 4AC \leqslant 0$ then $Ax^2 + Bx + C \geqslant 0$, so $(A \pm a)x^2 + (B \pm b)x + (C \pm c) \geqslant 0$ for all x, with $A \pm a \geqslant 0$. Their discriminants are $\leqslant 0$; adding the two inequalities gives $b^2 - 4ac \leqslant 4AC - B^2$.

Hint 4. For the other inequality (only needed if $4ac - b^2 > 0$), evaluate at the vertex x_0 of $Ax^2 + Bx + C$: $\frac{4ac-b^2}{4A} \leqslant \frac{4ac-b^2}{4|a|} \leqslant |ax_0^2 + bx_0 + c| \leqslant \frac{4AC-B^2}{4A}$.

Problem 10

Hint 1. First check the linear polynomials: $p(x) = x + c$ works ($q = 2/c$ for $c \neq 0$, any q for $c = 0$), and so does $p(x) = -x + c$, since then $p(p(x)) = x$. For the general case pull one factor $p(x) - x$ out of $p(p(x)) - x = \big(p(p(x)) - p(x)\big) + \big(p(x) - x\big)$, using that $p(y) - p(x)$ is divisible by $y - x$.

Hint 2. Write $p(y) - p(x) = (y - x)S(x, y)$ with a polynomial S; note $S(x, x) = p'(x)$. Then $p(p(x)) - x = \big(p(x) - x\big)\big(1 + S(x, p(x))\big)$, so for $p(x) \neq x$ the condition is $p(x) - x \mid 1 + S(x, p(x))$. Modulo $p(x) - x$ you may replace $p(x)$ by x, so the condition becomes

$$p(x) - x \;\mid\; p'(x) + 1.$$

(In terms of roots: every root of $p(x) - x$ must be a root of $p(p(x)) - x$ of at least twice its multiplicity.)

Hint 3. Compare degrees. If $\deg p = d \geqslant 2$, then $\deg(p(x) - x) = d > \deg(p' + 1)$, so $p' = -1$, which is impossible. For $p(x) = ax + b$: if $a = 1$ the divisor is a constant; if $a \neq 1$, the linear polynomial $(a - 1)x + b$ divides the constant $a + 1$ only when $a = -1$ (so a constant p, with $a = 0$, fails). Answer: $p(x) = x + c$ and $p(x) = -x + c$, $c \in \mathbb{R}$.